

4DVST

France is Warming Up

A story written in degrees

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2025-2026

Data Visualization | Météo France Open Data |
SUPINFO



Context

- ✔ **+2.1°C warming over last century:**
France has warmed significantly, with acceleration of warming trends since the 1980s.
- ✔ **Météo France observations:**
Daily data from 831 weather stations across France since 1950.
- ✔ **Data-driven storytelling:**
Project transforms raw meteorological data into a compelling visual narrative.

Objectives

- 🎯 **Prove warming trend:**
Demonstrate temperature rise with real data from 1950–2024.
- 🎯 **Regional comparison:**
Analyse climate patterns across 8 French regions and highlight extreme events: 2003, 2019, 2022, 2024.
- 🎯 **Interactive dashboard:**
Build Metabase dashboard to analyse precipitation patterns and drought trends.

France is Warming Up

Data Sources

Comprehensive climate data from Météo France Open Data powers this analysis, covering 75 years of daily weather observations across France.

Primary Dataset

Details

Daily observations 1950–2024

Météo France Open Data
Variables: Temperature (min/max/avg)
Precipitation, Wind, Frost

Geographic Coverage

Regions

8 representative departments selected

Nice (06), Marseille (13)
Bordeaux (33), Lille (59)
Strasbourg (67), Lyon (69), Paris (75)

Data Volume

Scale

8.6 million rows in PostgreSQL

~75 years of observations
8 departments × 831 stations
Full historical coverage

TECHNICAL APPROACH

Data Processing Pipeline

*Infrastructure: Docker Compose | Single command deployment

Météo France API

CSV + REST

Step 1: Data Retrieval

- Daily observations
- 8 departments
- 831 stations

Python Pipeline

10x Faster

Step 2: ETL Processing

- pandas + SQLAlchemy
- COPY PostgreSQL

Metabase Dashboard

15 Visuals

Step 4: Visualization

- 4 tabs + GeoJSON
- Interactive filters

PostgreSQL 15

8.6M Rows

Step 3: Database

- Materialized views
- Optimized indexes

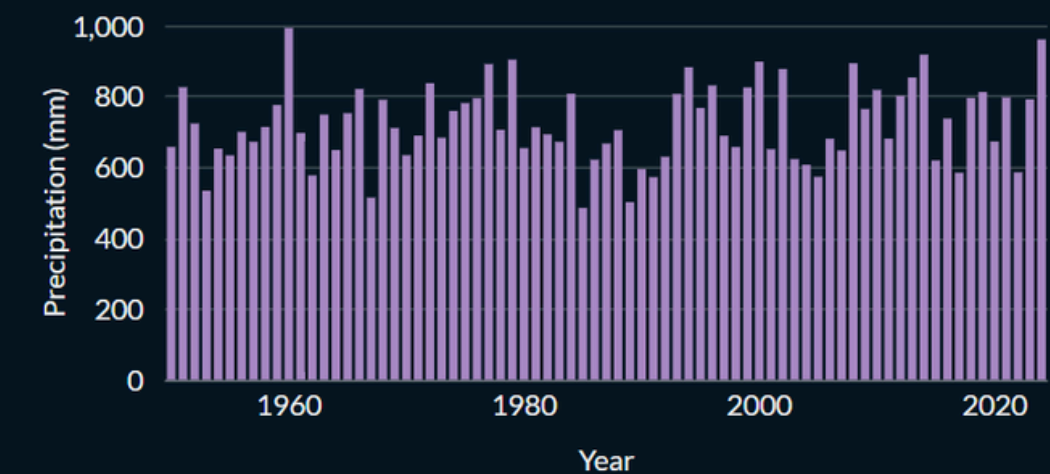
The Warming Story

France is Warming Up

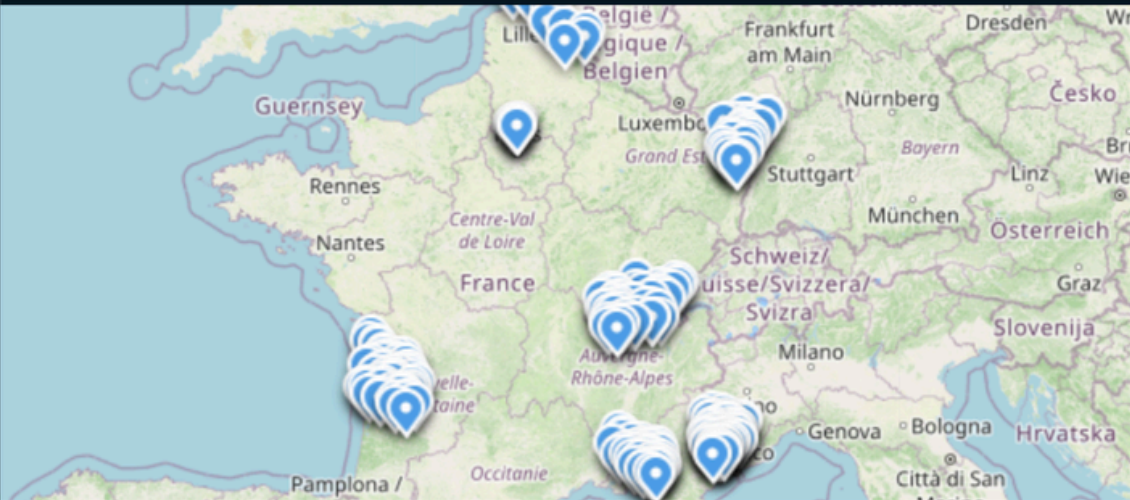
Warming Story Regional Comparison Extreme Events Precipitation

Year

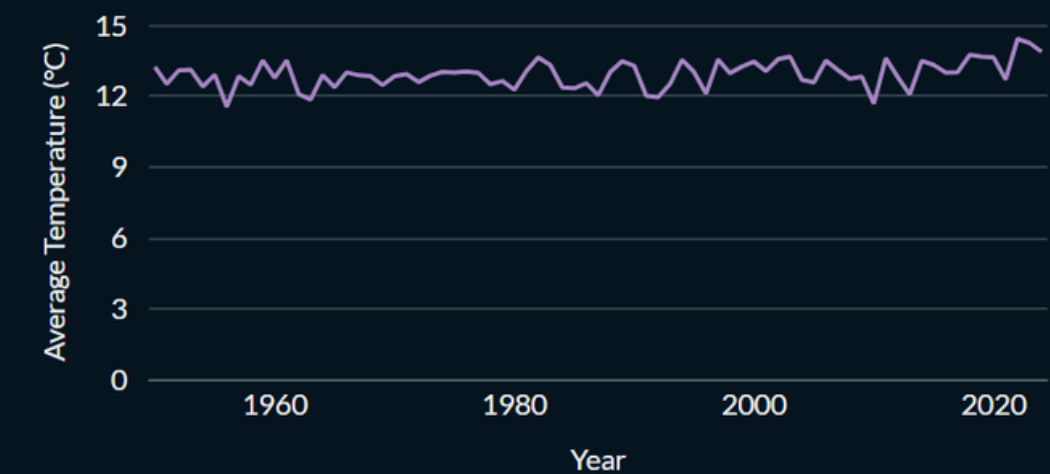
Annual Precipitation 1950-2024



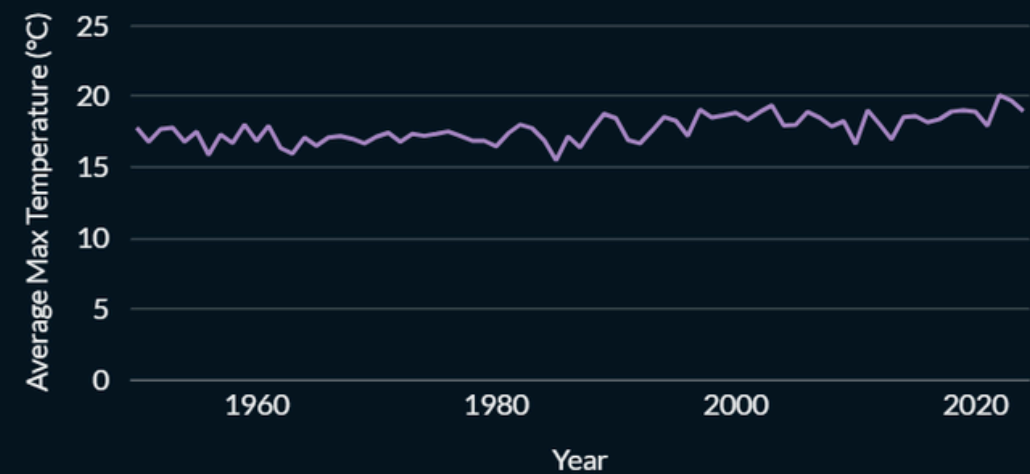
Station Temperatures Map



Temperature Trend 1950-2024



Max Temperature Records 1950-2024



Key Insight

Hottest years all after 2010



Top 6

Starting Point 1950 baseline



~12°C

Peak Years Extreme heat events



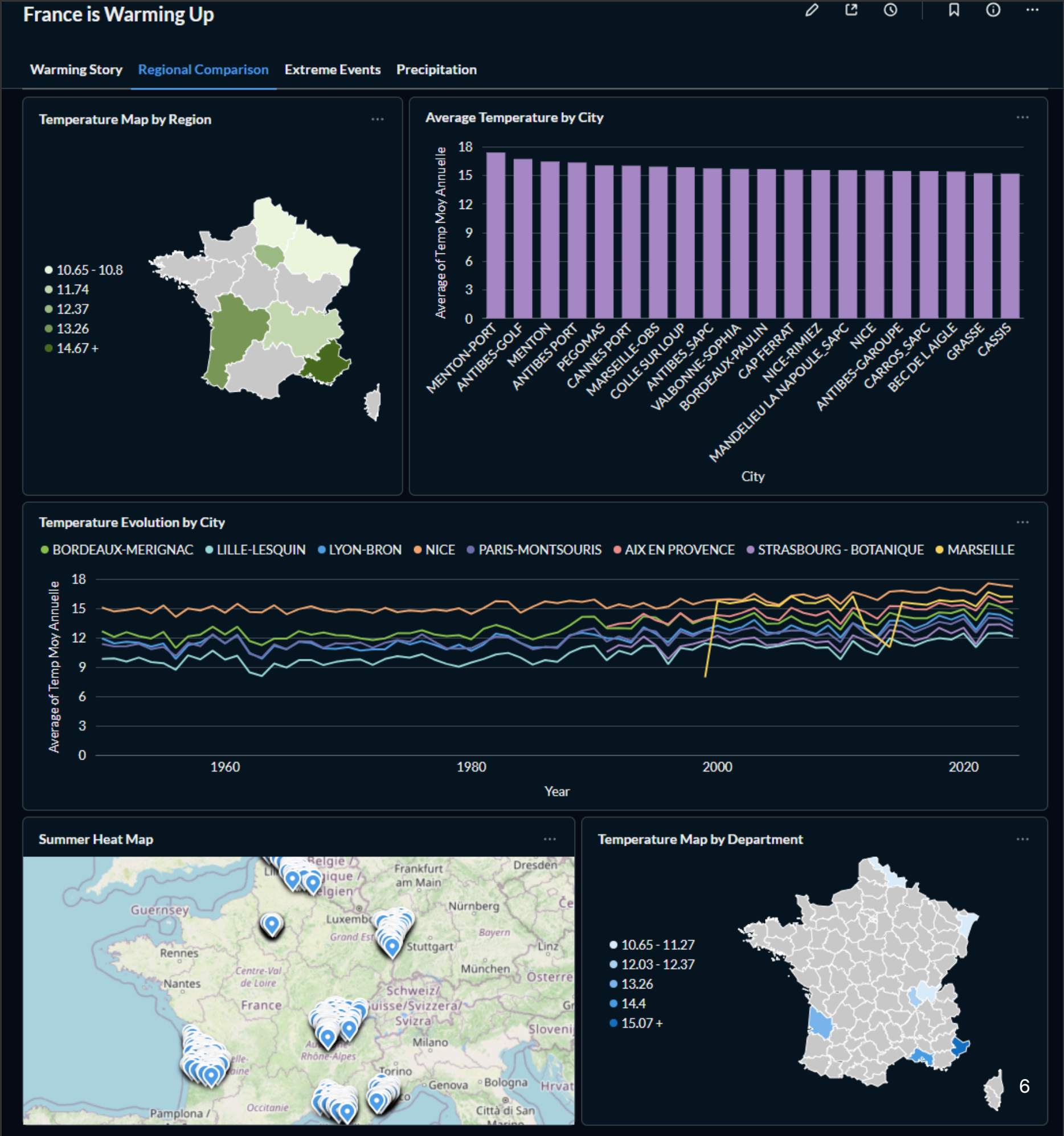
2003
2019
2022
2024

Regional Differences

All cities warming post-2000.
Nice and Marseille show fastest absolute warming.
Summer heat spreading northward.

Mediterranean coast is 6°C warmer than Northern France

*Based on Météo France data 1950–2024



EXTREME EVENTS

Major Heat Events in France*

2003 — Historic Heatwave

15,000

Deaths recorded

- National wake-up call
- Emergency response created

2019 — Absolute Record

44.6°C

At Gallargues-le-Montueux

- 20 departments on alert
- New national maximum

2024 — Threshold Crossed

+1.5°C

Above Paris Agreement limit

- First year exceeding target
- Global climate signal

2022 — Hottest Year

14.4°C

Annual average temperature

- Severe droughts
- Massive wildfires

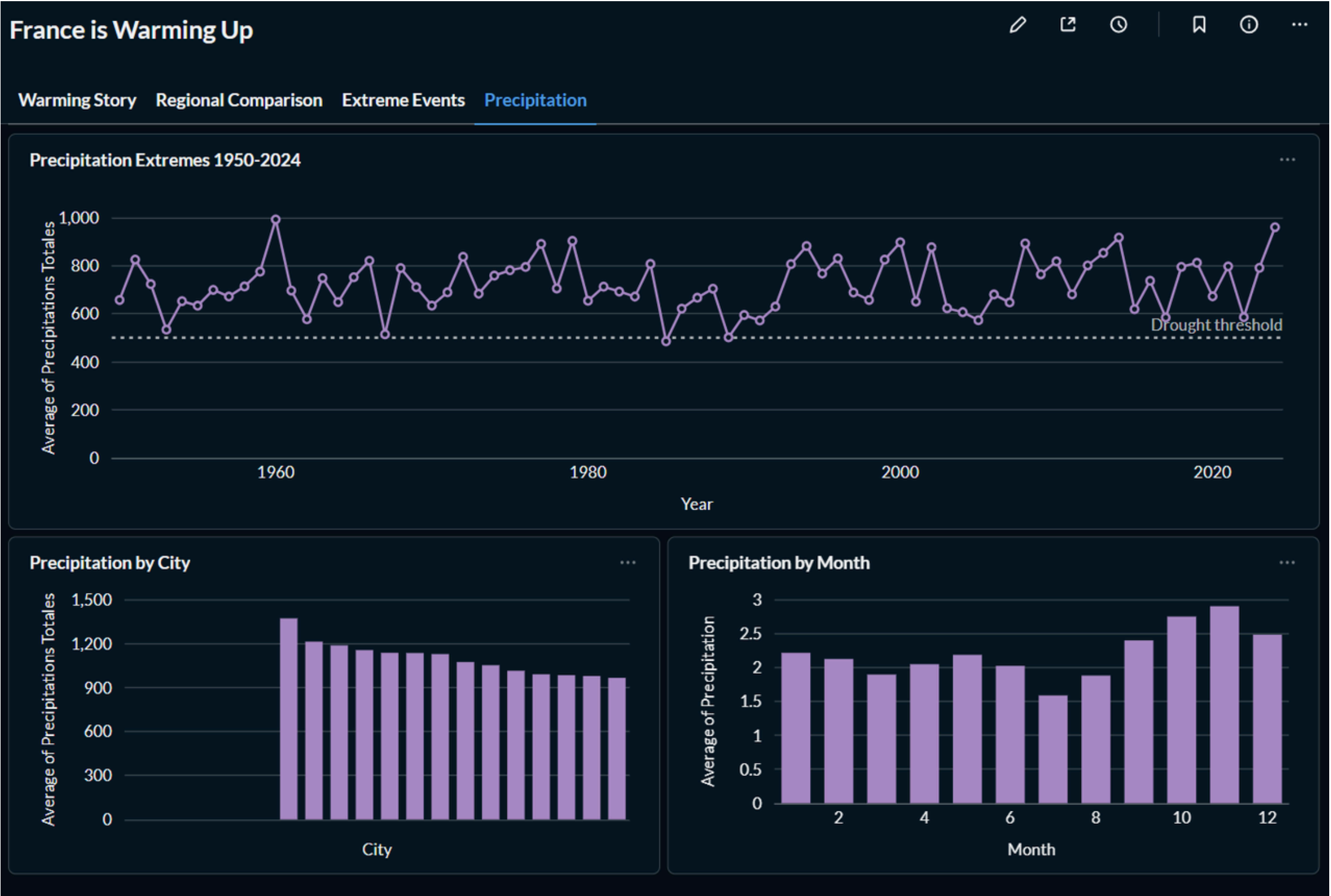
Precipitation

Patterns & Drought Analysis

Climate change creates a paradox: overall drier conditions alongside more intense precipitation events, with droughts and extreme rainfall now coexisting.

- ➡ **Summer Driest Season**
Months 6–7 consistently show lowest rainfall levels across all stations.
- ➡ **Autumn/Winter Peak Rainfall**
Most precipitation concentrated in late year, increasing variability since 2000.
- ➡ **2024 Near-Record Total**
Extreme precipitation peaks multiplying while drought periods intensify.

*Annual precipitation trends 1950–2024



DASHBOARD

Interactive Dashboard — France is Warming Up

Warming Story

Tab 1

Temperature trend 1950–2024

- Station temperatures map
- Annual precipitation
- Max temperature records
- Temperature Trend 1950-2024

Regional Comparison

Tab 2

Choropleth maps (GeoJSON)

- Temperature Map by Region
- Average Temperature by City
- Temperature Evolution by City
- Summer Heat Map
- Temperature Map by Department

Precipitation

Tab 4

Annual extremes + drought line

- Precipitation by city
- Monthly patterns
- Precipitation Extremes 1950-2024

Extreme Events

Tab 3

Hot days per year >35°C

- Frost days per year
- Wind speed trend
- Monthly heatmap
- Top 10 hottest years
- Monthly Temperature Heatmap

France is Warming Up

Agile Methodology

- ✔ Sprint 1 — Exploration:
Data exploration, storytelling definition, and dataset selection from Météo France sources.
- ✔ Sprint 2 — Infrastructure:
Docker setup, PostgreSQL installation, Metabase configuration, and GitHub repository.
- ✔ Sprint 3 — Data Pipeline:
Python ETL scripts, COPY PostgreSQL method for 8.6M rows, data quality report.

SCRUM Project Workflow

- ✔ Sprint 4 — Dashboard:
4 tabs built, 15 visualizations, GeoJSON maps, and interactive filters.
- ✔ Sprint 5 — Presentation:
PDF slides, demo rehearsal, final delivery, and peer review.
- ✔ Project Management:
5-week project with daily standups, GitHub version control, and iterative delivery.

This story is not over.

France is Warming Up Even More

"Limiting global warming remains essential to shaping tomorrow's climate. Every tenth of a degree counts."
— Météo France, 2025



+2.1°C

France warmed in 1
century



44.6°C

Absolute record 2019



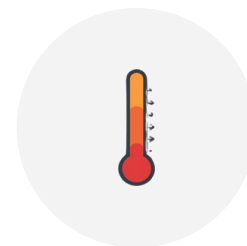
+2.7°C

Projected by 2050

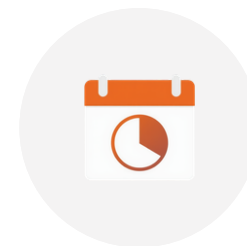


x10

More hot days since
1980



Data: meteo.data.gouv.fr ·
Météo France Open Data



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